

Customer Segmentation with K-Means

Segmenting 3,900 retail customers, and testing whether the segments are real

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Code and notebook: github.com/RatanaSovann/customer-segmentation

1. Executive summary

This project groups the 3,900 customers in the Customer Shopping Trends dataset into segments a marketing team could target, using K-Means clustering on age, purchase amount, previous purchases and review rating.

- Four evenly sized segments come out of the model and are easy to describe, for example *younger*, *higher spend*, *dissatisfied*. Each one has a clear recommended action.
- Validation shows the segments are **not natural groups**. The silhouette score on the real data matches uniform random noise at every number of clusters, different random seeds give different segments (median ARI 0.47), and the segments don't differ on subscription, discount use or product category.
- The exploratory analysis agrees. Once raw counts are converted to shares, gender, subscription status, age and spend level don't predict what or how often customers buy.
- **Recommendation:** use the segments as a simple age x spend x satisfaction targeting grid, not as discovered customer types, and rerun the pipeline on real, larger transaction data before relying on it.

2. Business problem

A retailer wants to understand its customers well enough to tailor marketing: who spends more, who is unhappy, and which groups should get which offers. The two goals are to (1) identify customer segments from purchasing and demographic data and (2) turn them into targeting recommendations.

3. Data

The [Customer Shopping Trends dataset](#) (Kaggle) is synthetic. It has 3,900 rows (one purchase per customer) and 18 columns, with no missing values or duplicates.

Group	Columns
Demographics	Age (18-70), gender, location (US state)
Purchase	Item (25 products), category (4), amount (USD 20-100), size, colour, season
Behaviour	Review rating (2.5-5.0), previous purchases (1-50), purchase frequency, subscription, discount, promo code, shipping type, payment method

The customer base is uneven: 68% of customers are men and 73% don't subscribe. This matters for the exploratory analysis below.

4. Exploratory analysis

Comparisons use **shares within each group** rather than raw counts, and a chi-square test of independence checks whether each difference is bigger than chance.

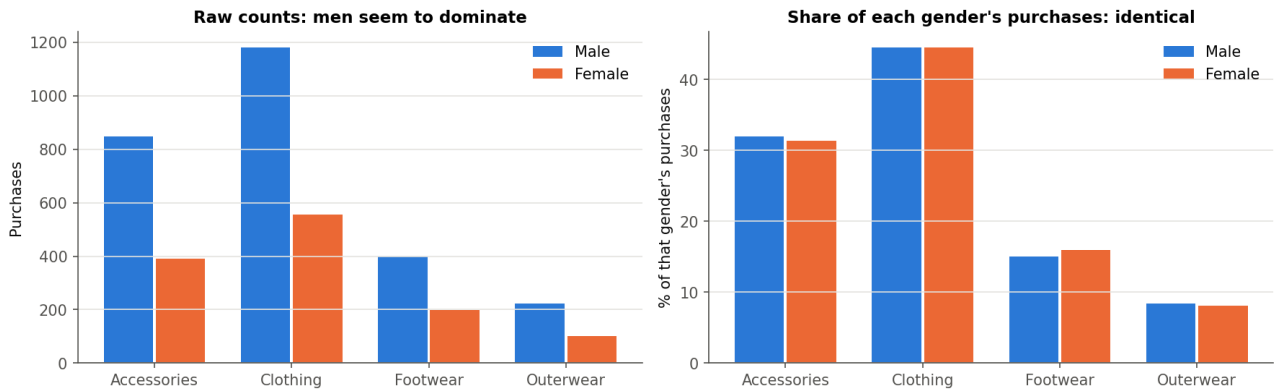


Figure 1. Raw counts suggest men dominate every category, but that's only because 68% of customers are men. Within each gender the category mix is identical ($p = 0.90$).

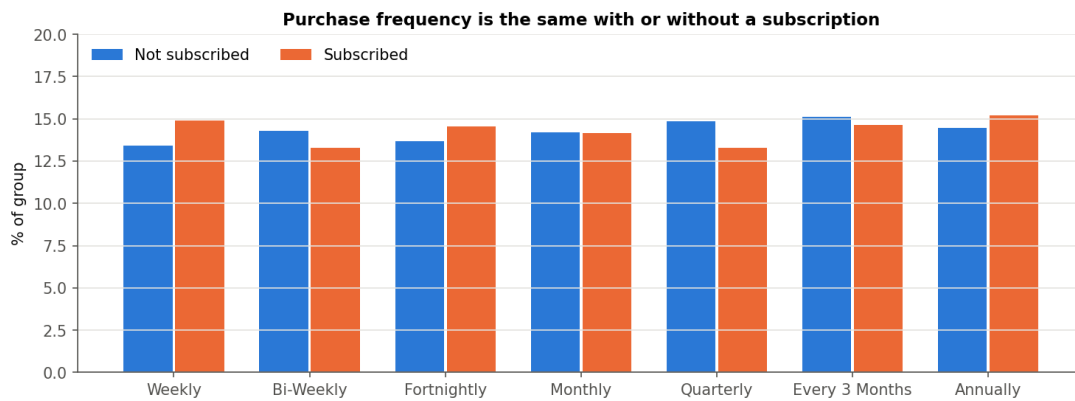


Figure 2. Subscribers and non-subscribers buy at the same frequencies ($p = 0.69$).

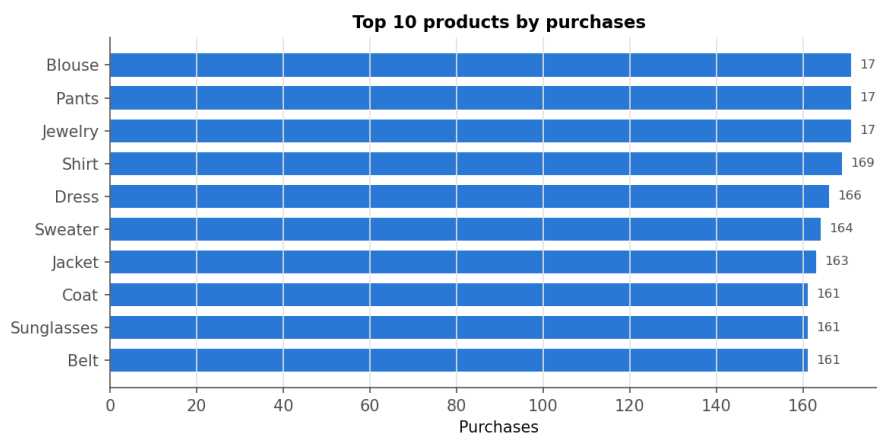


Figure 3. All 25 products sell in similar volumes, from 124 (jeans) to 171 (blouse, pants, jewelry).

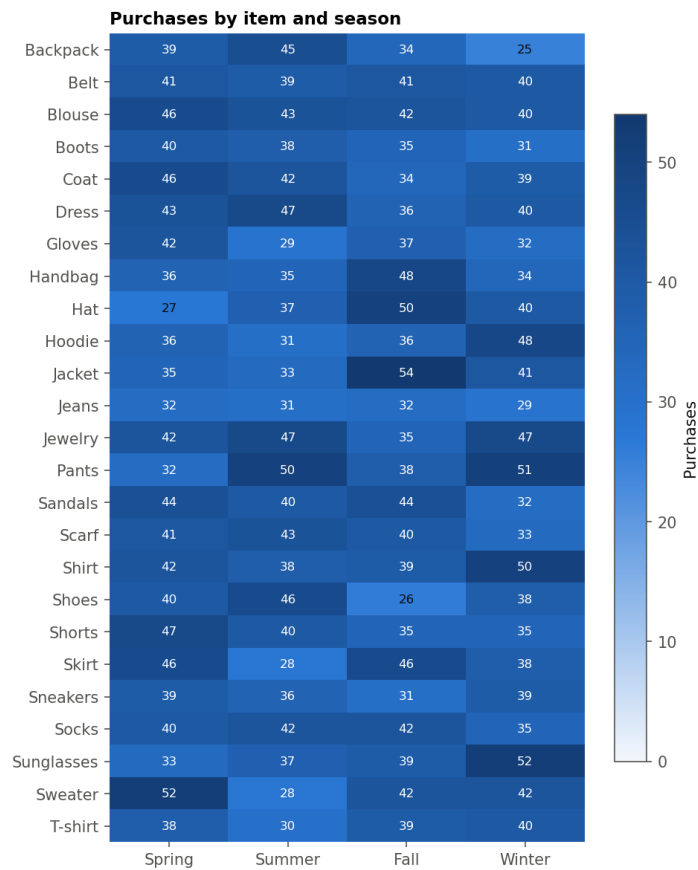


Figure 4. Purchases by item and season range from 25 to 54 per cell (about 39 expected), which is consistent with chance ($p = 0.29$). There is no reliable seasonal pattern.

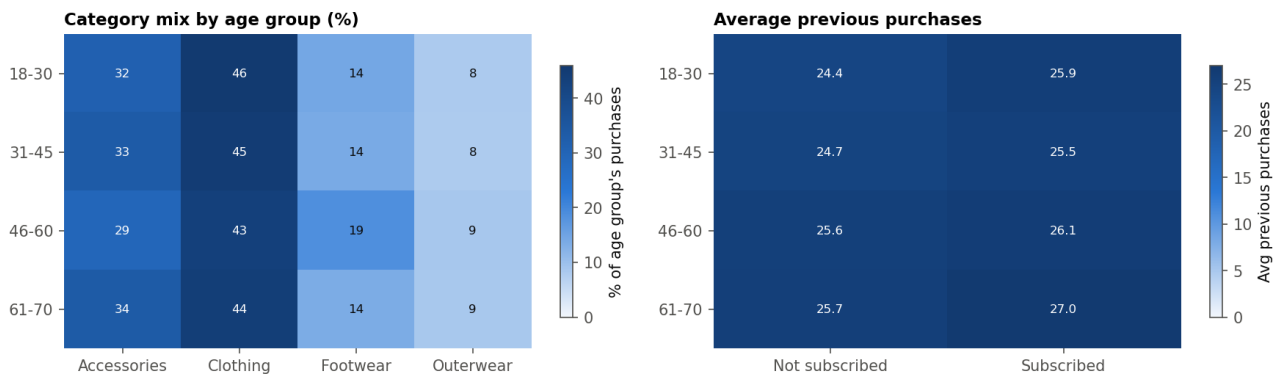


Figure 5. Category mix is similar across age groups. The only borderline difference is that 46-60 year-olds buy more footwear (19% vs about 14%, $p = 0.053$). Subscribers have slightly more previous purchases in every age group (+1 to +1.5).

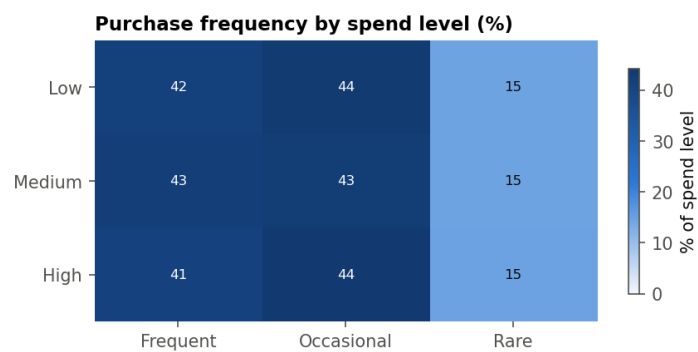


Figure 6. Purchase frequency is the same at every spend level ($p = 0.96$).

Chi-square tests

Comparison	p-value
Item purchased x season	0.29
Category x gender	0.90
Purchase frequency x subscription	0.69
Category x age group	0.053
Purchase frequency x spend level	0.96

None of the tests is significant at the 5% level. Across five tests, one near-miss such as age x category is about what chance alone would produce.

5. Methodology

Features

K-Means uses Euclidean distance, so only numeric features are used: **age, purchase amount, previous purchases and review rating**. Each is standardised to mean 0 and SD 1 so that no feature dominates the distance. Purchase amount (dollars) and previous purchases (a count) are kept as separate features, because adding them together would produce a number with no clear unit.

All four numeric features are close to uniform

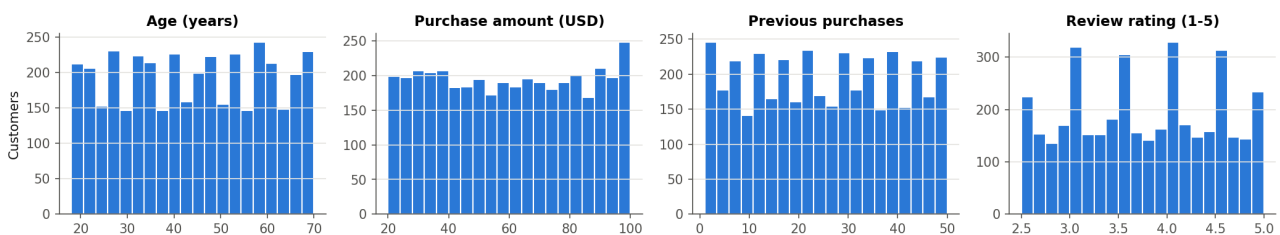


Figure 7. All four numeric features are close to uniform, and their pairwise correlations are near zero.

Choosing k

I compared $k = 2$ to 8 using inertia (the elbow method) and the silhouette score. I also ran the same pipeline on **uniform random data with the same shape**. If the real data scores no better than noise, the clusters are cutting up an even cloud of points rather than finding groups.

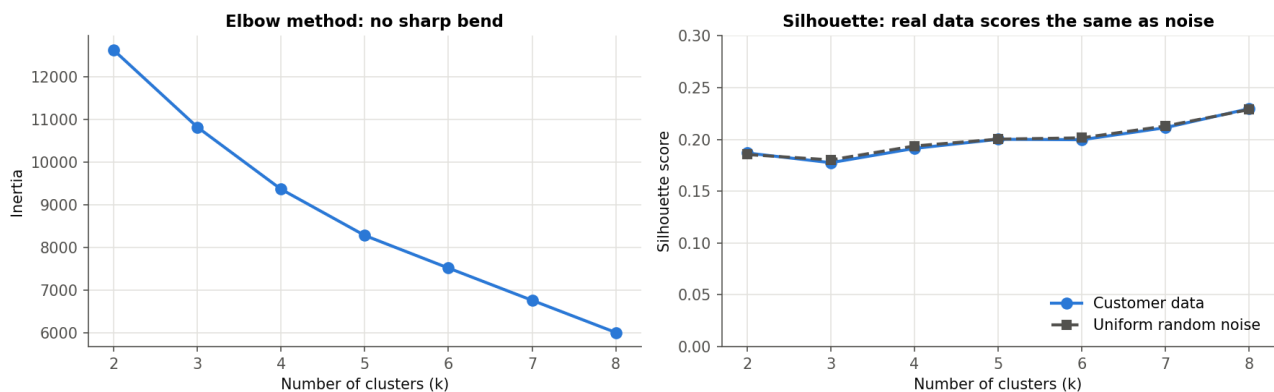


Figure 8. There is no elbow, and the silhouette score (0.18-0.23) tracks the noise baseline at every k .

Because the data has no natural cluster count, I chose **$k = 4$** for interpretability: four segments are easy to act on, and $k = 4$ is a small local peak in silhouette over $k = 3$.

Validation

- **Stability:** refit with 10 random seeds and compare assignments using the Adjusted Rand Index (1 = identical, 0 = chance).
- **External validity:** compare segments on variables that weren't used for clustering (subscription, discounts, gender, category).

6. Results

The four segments

#	Segment	Age	Spend	Rating	Customers	Suggested action
0	Older, higher spend, satisfied	56	\$82	4.0	966	Protect and reward: VIP perks, early access, referral incentives
1	Younger, lower spend, satisfied	31	\$55	4.4	969	Grow spend: bundles, cross-sell, personalised upsell offers
2	Younger, higher spend, dissatisfied	33	\$65	3.1	986	Most at-risk revenue: follow up on reviews, service recovery
3	Older, lower spend, dissatisfied	56	\$38	3.5	979	Re-engage cheaply: feedback surveys, targeted win-back offers

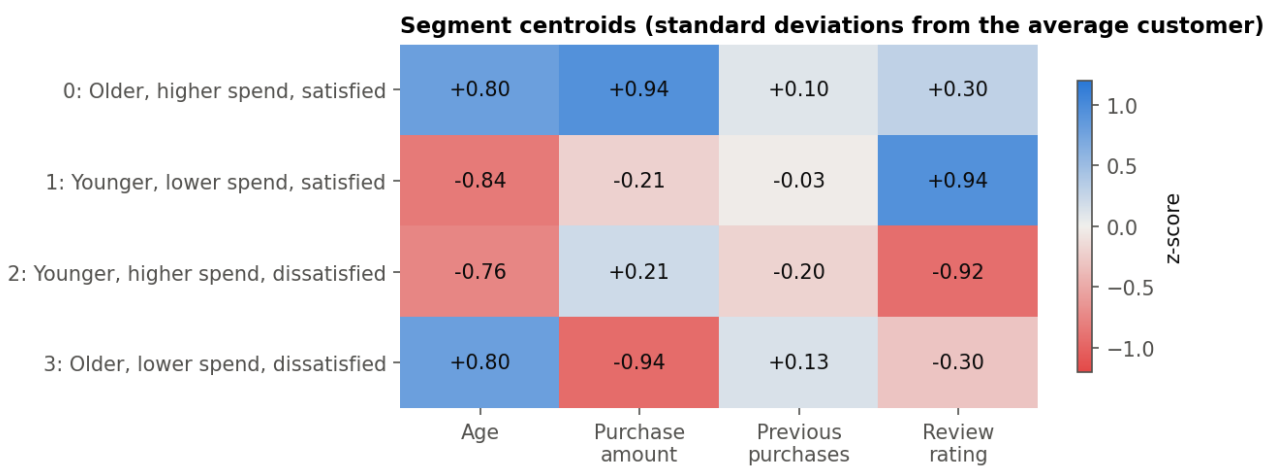


Figure 9. Standardised segment centroids. Segments are separated by age, spend and review rating. Previous purchases hardly varies (within +/-0.2 SD of average).

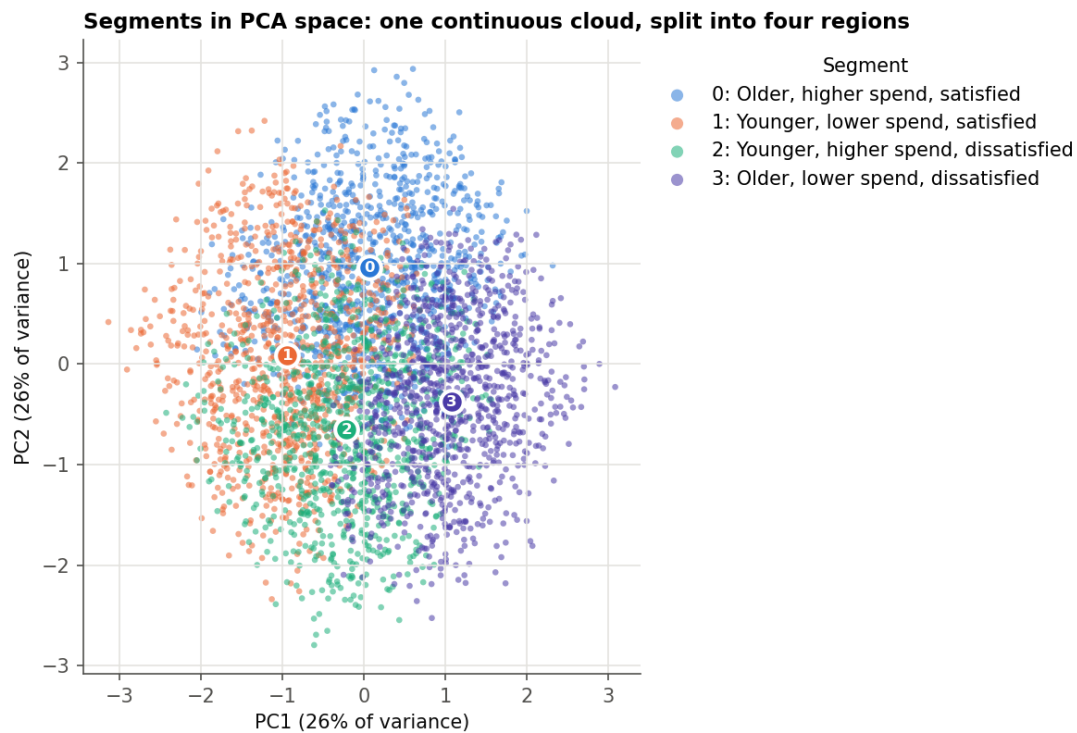


Figure 10. The segments projected onto two principal components form one continuous cloud divided into regions, not separate groups.

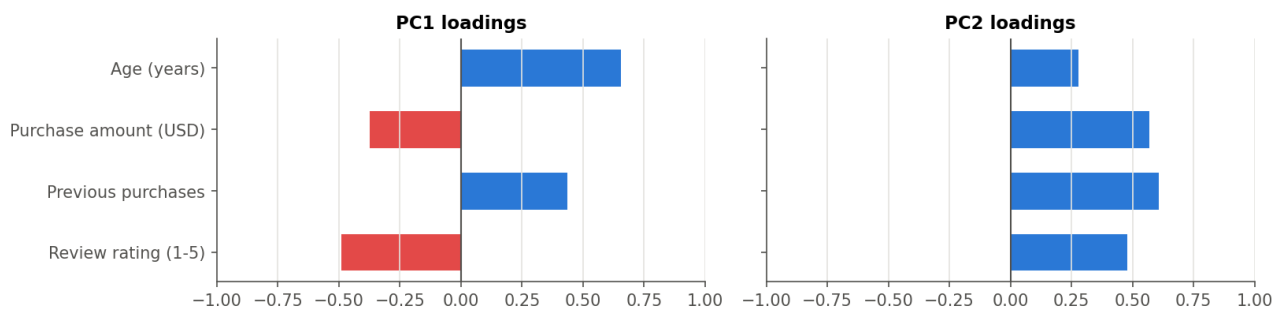


Figure 11. PC1 contrasts older customers with more previous purchases against younger, higher-spend, higher-rating ones. PC2 weights all four features positively.

Are the segments real?

- **Silhouette vs noise:** identical at every k (Figure 8).
- **Stability:** median ARI across seeds is **0.47**. The data can be split in several near-equal ways, and the random start picks one of them.
- **External validity:** subscription rate (26-28%), discount use (42-45%), gender mix and category mix are almost the same in every segment.

All three checks agree: K-Means always returns k clusters, even when the data has none. Checking against a noise baseline catches this before the segments reach a stakeholder.

7. Recommendations

- **Use the segments as a targeting grid** on age, spend and satisfaction, with the actions in the results table. Prioritise segment 2 (younger, higher spend, dissatisfied), which has the most revenue at risk.
- **Consider plain rules instead.** On this data, splitting by median age, spend and rating gives similar groups that are easier to explain and maintain.
- **Don't target by gender or subscription status alone.** Neither predicts category preference or purchase frequency here.
- **Test before scaling.** Run A/B tests on segment-specific offers to measure lift, instead of assuming the segments behave differently.

8. Limitations

- **The dataset is small.** 3,900 customers, each with a single transaction and no dates, so behaviour over time can't be studied. Cross-tabs get thin fast (about 39 purchases per item-season cell), which limits the power to detect small effects, and the findings may not generalise to a larger or real customer base.
- **The data is synthetic.** Its features are independent and evenly spread, so no clustering method could find natural groups in it. The pipeline is ready to run on real transaction data.
- **Categorical features are left out.** K-Means needs numeric inputs, so category, season and payment method aren't used. K-Prototypes or Gower distance with hierarchical clustering could include them.
- **Fairness.** Segments based on age should be reviewed before use, so that offers don't systematically exclude groups that are under-represented in the data.

9. Next steps

- Apply the pipeline to real, multi-period transaction data and build RFM-style (Recency, Frequency, Monetary) features.
- Try K-Prototypes or Gower distance so categorical behaviour (category, season, payment) can shape the segments.
- Keep the noise baseline and stability checks as a standard gate before any segmentation is shared.

References

Customer Shopping Trends Dataset, Kaggle.

<https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset>

Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53-65.

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